

Test 1

- Every solution MUST be written using complete sentences. Include all necessary reasoning in a correct order so as to construct a complete and coherent logical argument.
 - You may use any definition/theorem we did in class (clearly specify what result you are using), unless the question asks you to prove something standard. Ask me in case of any doubt.
 - Upload your answers on moodle. Separate pdf/jpeg file for each problem.
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0. Let $f : V \rightarrow W$ be a linear map between finite dimensional vector spaces.

(a) Show that the image $f(V)$ is a subspace of W .

(b) Let $S = \{z_1, \dots, z_\ell\} \subset V$. For each statement below, state if it is true (and prove it) or false (and give a simple counterexample).

1. If S is linearly independent, then $f(S)$ is linearly independent.
2. If $f(S)$ is linearly independent, then S is linearly independent.
3. If S spans V , then $f(S)$ spans $f(V)$.
4. If $f(S)$ spans $f(V)$ then S spans V .

(c) Complete the following (slightly different) proof of the rank-nullity theorem for f .

Let $w_1 = f(v_1), \dots, w_r = f(v_r)$ be a basis for $f(V)$, i.e., a basis for the image of f .

Claim: The vectors _____, together with a basis $x_1, \dots, x_n$ of _____, form a basis of _____.

Proof of spanning: Consider an arbitrary vector y in _____. Then ...

Proof of independence: Suppose there is a linear relation _____. Then ...

1. Consider the linear map $f : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ given $f(x) = Mx$ where M is a 4×3 matrix. M is not given but you are told that doing a series of column operations $op_1, op_2, \dots, op_7$ to M gives the

matrix $N = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 4 \\ 20 & 30 & 40 \end{bmatrix}$. Note that one can recover M from N using the operations .

Give an ordered basis $\mathbf{B}$ of $\mathbb{R}^3$ and an ordered basis $\mathbf{C}$ of $\mathbb{R}^4$ such that each entry of the matrix of f with respect to these new bases is 0 or 1, with the fewest 1's possible.

Your answer must be of the following form. For each basis, if possible, give a list of vectors. If the given information is not enough to write actual vectors, then give an algorithmic procedure in terms of the given information. Justify.

2. Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be linear maps between finite dimensional vector spaces.

Prove the four inequalities below. Here r stands for the rank (i.e., dimension of the image) of a linear map and n stands for its nullity (i.e., dimension of the kernel). Possibly useful considerations: $T(U)$ is a subspace of V . By restricting S to this subspace, we have the linear map $S|_{T(U)} : T(U) \rightarrow W$.

(a) $r(ST) \leq r(S)$. (b) $n(T) \leq n(ST)$. (c) $r(ST) \leq r(T)$. (d) $n(ST) \leq n(S) + n(T)$.

3. (a) For v in $\mathbb{R}^2$, define $F(v)$ to be the reflection of v in the line $y = 5x$. Find the matrix of the linear operator F with respect to the standard basis. Then find a basis w_1, w_2 of $\mathbb{R}^2$ with respect to which the matrix of F is diagonal.

(b) Show that any 2×2 matrix A satisfying $A^2 = \text{the identity matrix } I$ is either $\pm I$ or conjugate to the matrix of a reflection in some line, i.e., of the form $P^{-1}(\text{the matrix of a reflection})P$.

4. Short independent questions

(A) For an $r \times c$ matrix A , consider the functions $f(x) = Ax$ and $g(y) = A^T y$. Suppose g is a surjective function. What does that imply about the function f ? Give the strongest answer that you can. Is the converse of your implication true?

(B) You are given a set of vectors $v_1, \dots, v_k$ in some $\mathbb{R}^n$. State how you will solve the following. Your answer must be completely algorithmic. You may assume as black box ONLY row reduction and its connection with matrix multiplication. You may NOT say “solve this equation” or “invert this matrix”. If that’s what you need to do, you must actually state a programmable procedure, including precise instructions about how you will make any required decision.

1. Are $v_1, \dots, v_k$ linearly independent?
2. Do $v_1, \dots, v_k$ span $\mathbb{R}^n$?
3. Assume $k = n$ and that $v_1, \dots, v_n$ form a basis of $\mathbb{R}^n$. For a given vector b , how will you find the coordinates of b in terms of this new basis?

(C) Prove multiplicativity of the determinant function as follows. For a fixed $n \times n$ matrix A define the function $f(AX) = \det(AX) = \det(Ax_1, Ax_2, \dots, Ax_n)$ where X is a square matrix with column vectors $x_1, \dots, x_n$. You may use that determinant is the unique function such that ...

(D) Let $A = \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix}$, $B = \begin{bmatrix} b_1 & 0 & 0 \\ 0 & b_2 & 0 \\ 0 & 0 & b_3 \end{bmatrix}$ and $M_{2 \times 3}$ = the vector space of 2×3 matrices.

For any X in $M_{2 \times 3}$, define $T(X) = AX - XB$. Let M be the matrix of the linear map T with respect to some bases.

- Show that $\det(M) = 0$ if and only if $\{a_1, a_2\} \cap \{b_1, b_2, b_3\}$ is nonempty.
- Calculate $\text{rank}(T)$.